Mini Project - Healthcare Analytics (PowerBI & Tableau)

Code:

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Treatment Cost Breakdown by Hospital (Grouped Bar Chart)
cost_df = pd.read_csv("hospital_cost_data.csv")
cost_df = cost_df.set_index('Hospital')

fig, axes = plt.subplots(3, 2, figsize=(15, 18))
plt.subplots_adjust(hspace=0.4)

cost_df.plot(kind='bar', ax=axes[0, 0], colormap='tab10', width=0.8)
axes[0, 0].set_title('Treatment Cost Breakdown by Hospital')
axes[0, 0].set_ylabel("Avg Cost per Patient (INR '000)")
axes[0, 0].tick_params(axis='x', rotation=45)

# 2. Monthly Patient Admissions Trend (Emergency vs Planned)
admissions = pd.read_csv("monthly_admissions.csv")
axes[0, 1].plot(admissions['Month'], admissions['Emergency Admissions'],
        marker='o', color='firebrick', label='Emergency')
axes[0, 1].plot(admissions['Month'], admissions['Planned Admissions'],
        marker='^', linestyle='--', color='seagreen', label='Planned')
axes[0, 1].set_title('Monthly Patient Admissions Trend')
axes[0, 1].set_xlabel('Month')
axes[0, 1].set_ylabel('Number of Admissions')
axes[0, 1].set_xticks(range(1, 13))
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axes[0, 1].legend()
axes[0, 1].grid(alpha=0.3)

# 3. Length of Stay Distribution by Department (Violin Plot)
patients = pd.read_csv("patient_records.csv")
sns.violinplot(x='Department', y='Length of Stay (Days)', data=patients,
            hue='Department', legend=False, inner='quartile', cut=0,
            palette='Set2', ax=axes[1, 0])
axes[1, 0].set_title('Length of Stay Distribution by Department')
axes[1, 0].tick_params(axis='x', rotation=30)

# 4. Inter-Department Patient Referrals (Heatmap)
referral_df = pd.read_csv("referral_data.csv")
pivot = referral_df.pivot_table(index='From Department',
                        columns='To Department', values='Referrals',
                        aggfunc='sum', fill_value=0)
sns.heatmap(pivot, annot=True, fmt="d", cmap="Blues", linewidths=0.5,
          ax=axes[1, 1])
axes[1, 1].set_title('Inter-Department Patient Referrals')

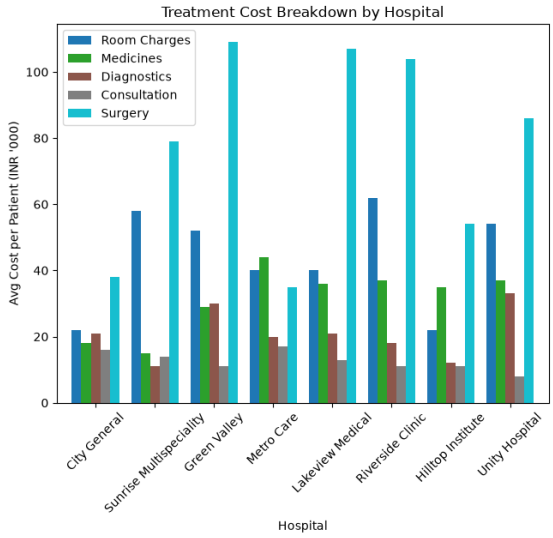
# 5. Patient Count by Department (Horizontal Bar Chart)
dept_counts = patients['Department'].value_counts().sort_values()
dept_counts.plot(kind='barh', ax=axes[2, 0],
              color=sns.color_palette('crest', len(dept_counts)),
              edgecolor='black')
axes[2, 0].set_title('Patient Count by Department')
axes[2, 0].set_xlabel('Number of Patients')
axes[2, 0].set_xlim(0, dept_counts.max() * 1.12)
for i, v in enumerate(dept_counts):
    axes[2, 0].text(v + 2, i, str(v), va='center', fontweight='bold')

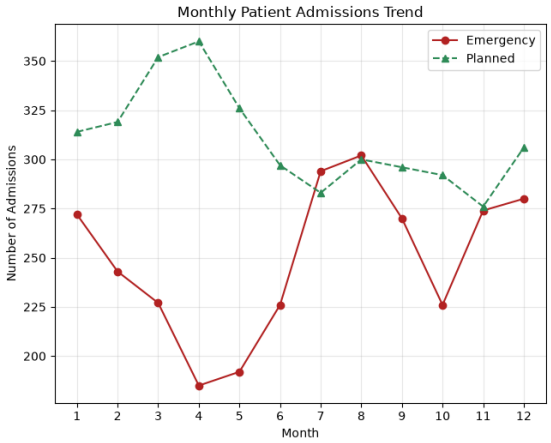
# 6. Length of Stay vs Bill Amount (Regression Scatter Plot)
sns.regplot(x='Length of Stay (Days)', y="Bill Amount (INR '000)",
          data=patients, ax=axes[2, 1],
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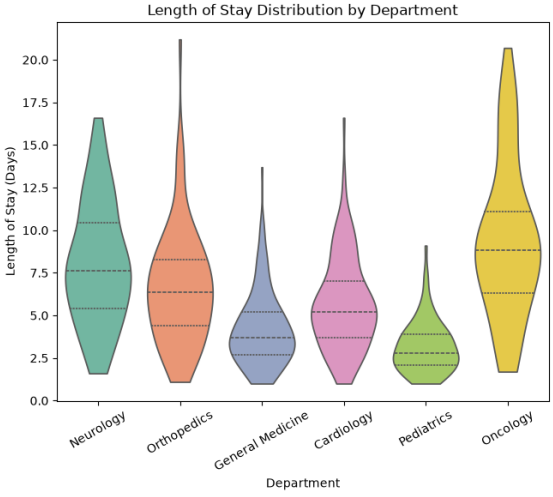
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        scatter_kws={'alpha': 0.4, 'color': 'darkorange'},
        line_kws={'color': 'navy'})
corr = np.corrcoef(patients['Length of Stay (Days)'],
                   patients["Bill Amount (INR '000)"])[0, 1]
axes[2, 1].set_title(f'Length of Stay vs Bill Amount (r = {corr:.2f})')

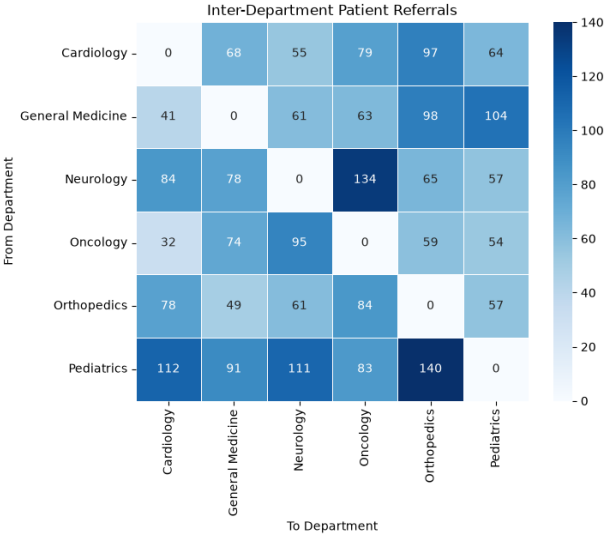
plt.tight_layout()
plt.savefig('Healthcare_Analytics_Dashboard.png')
print("Visualizations successfully saved to Healthcare_Analytics_Dashboard.png")
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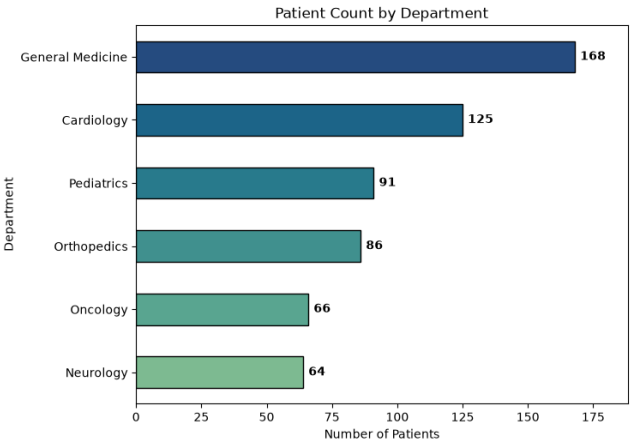
Output:

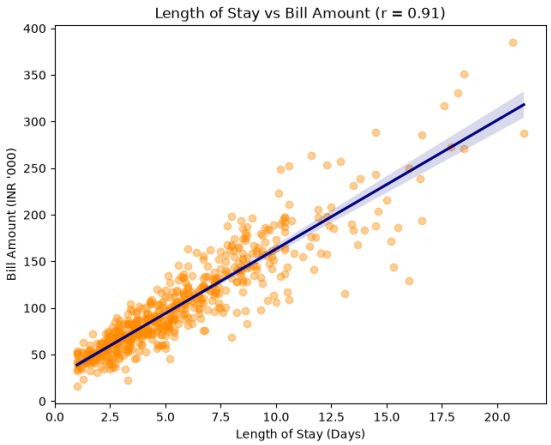

Treatment Cost Breakdown by Hospital
Room Charges
Medicines
Diagnostics
Consultation
Surgery
Avg Cost per Patient (INR '000)
City General
Sunrise Multispeciality
Green Valley
Metro Care
Lakeview Medical
Riverside Clinic
Hilltop Institute
Unity Hospital
Hospital


Monthly Patient Admissions Trend
Emergency
Planned
Number of Admissions
Month


Length of Stay Distribution by Department
Length of Stay (Days)
Neurology
Orthopedics
General Medicine
Cardiology
Pediatrics
Oncology
Department


Inter-Department Patient Referrals
Cardiology 0 68 55 79 97 64
General Medicine 41 0 61 63 98 104
Neurology 84 78 0 134 65 57
Oncology 32 74 95 0 59 54
Orthopedics 78 49 61 84 0 57
Pediatrics 112 91 111 83 140 0
From Department
To Department


Patient Count by Department
General Medicine 168
Cardiology 125
Pediatrics 91
Orthopedics 86
Oncology 66
Neurology 64
Department
Number of Patients


Length of Stay vs Bill Amount (r = 0.91)
Bill Amount (INR '000)
Length of Stay (Days)